Krish Maniyar

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Education

Vellore Institute of Technology (VIT) **2023 – 2027**
B.Tech, Computer Science and Engineering | ***CGPA: 9.27 / 10*** *Vellore, Tamil Nadu*

Experience

Marwadi Shares & Finance Ltd. (MSFL) **May 2025 – Jun 2025**
Flutter Application Development Intern *Rajkot, Gujarat*

- Developed the Flutter mobile frontend for **MChat**, a real-time chat application serving **150+ organizational users** (supporting **150+ concurrent connections**), integrating a .NET Web API and a **SignalR hub** for group/direct chats, read receipts, and presence tracking, reducing message latency by **40%** (verified by QA testing).
- Engineered **reliable SignalR connection management** with automatic reconnection, heartbeat/keep-alive pings, app pause/resume lifecycle hooks, and a **Long Polling fallback** to resolve WebSocket negotiation failures in the UAT environment.
- Built a secure **file management system** supporting multipart uploads up to **5 GB**, background download queues, and local gallery storage with platform permissions; debugged server-side SSL `CERTIFICATE_VERIFY_FAILED` errors caused by IP mismatches.

Centre for Social Entrepreneurship and Development (CSED), VIT **Apr 2024 – Feb 2025**
Core Committee Member *Vellore, Tamil Nadu*

- Organized a **national-level hackathon** with **200+ participants**, coordinating event logistics, sponsor engagement, and cross-functional teams, achieving a **60% increase** in event reach.
- Collaborated with student clubs, faculty, and external partners to plan and execute technical and entrepreneurship-focused events across the academic year.

Projects

Strategic News Analyzer | *FastAPI, PostgreSQL, pgvector, Redis, Celery, Next.js, Docker, HDBSCAN, RAG* GitHub

- Built a **decoupled microservices backend** (FastAPI + Celery Beat) ingesting news from **5+ sources** (NewsAPI, GNews, GDELT, RSS), deduplicating via **SHA-256 hashing**, running parallel AI analysis (sentiment, bias, strategic scoring) via `asyncio.gather()`, and generating **768-dim embeddings** with Ollama (`nomic-embed-text`).
- Designed a **hybrid RAG retrieval engine** combining **pgvector HNSW** ANN search and **PostgreSQL full-text search**, streaming LLM responses over **SSE**; clustered articles into geopolitical events using **HDBSCAN** on cosine-distance embeddings.
- Deployed with **Docker Compose** (FastAPI + Celery workers + Redis + PostgreSQL), exposed a **WebSocket live article feed** and Swagger API, with Supabase JWT-based authentication and Row-Level Security enforced across all database operations.

Query Weaver | *FastAPI, React, MySQL, ChromaDB, SQLAlchemy, Groq LLM, Sentence-Transformers, SSE* GitHub

- Built a **natural-language-to-SQL system** embedding user questions via `all-MiniLM-L6-v2` and performing **vector similarity search** in ChromaDB to retrieve the top-3 relevant tables; injected schema context, sample rows, and conversation history into **Groq (Llama 3.3-70B)** for SQL generation (end-to-end latency: ~**1.14s**).
- Implemented **SQL safety controls** detecting `INSERT`/`UPDATE`/`DELETE`/DDL and requiring explicit confirmation before execution; **auto-synchronizes schema changes** to ChromaDB to keep the vector store consistent with the live database.
- Streamed schema-change events to the React SPA via **Server-Sent Events**; instrumented per-stage timing — schema retrieval **44ms**, SQL execution **2–3ms**, LLM generation ~**1.02s** — via structured rotating logs.

Aneri Enterprise | *Next.js, TypeScript, Supabase (PostgreSQL), Nodemailer, Netlify* anerienterprise.shop

- Built and **deployed to production** a full-stack business website with a dynamic product catalog, secure driver download portal, and dual-channel enquiry system (Supabase DB + Gmail SMTP) serving real customers.
- Hardened the API with **IP-keyed rate limiting**, honeypot anti-spam, XSS/CRLF injection prevention, and an **SSRF-protected download proxy**; enforced **PostgreSQL RLS** restricting the anon key to SELECT on content tables and INSERT on analytics tables only.
- Built a **fire-and-forget analytics system** tracking site visits, product clicks, and driver downloads — all stored in Supabase with insert-only RLS; implemented **parallel data fetching**, session-based deduplication, and a 280ms debounced search to minimize redundant renders and DB writes.

Technical Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL
Backend & APIs: FastAPI, Flask, .NET Web API, RESTful APIs, SignalR, WebSockets, Server-Sent Events
Frontend & Mobile: React, Next.js, Tailwind CSS, Flutter (Dart)
Databases: PostgreSQL, MySQL, SQLite, ChromaDB, Redis, Supabase, Firebase
AI / ML: PyTorch, Hugging Face Transformers, LangChain, RAG, Sentence Transformers, Scikit-learn, NLP, MLflow
Cloud & DevOps: AWS (EC2, S3, IAM), Docker, Linux, Git, Uvicorn, Netlify, Vercel
Core CS: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

Certifications & Achievements

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate **Jan 2026**
Oracle University

Yantra Central Hackathon | ***Finalist*** **Feb 2025**
VIT Vellore